



CASUALTY ACTUARIAL SOCIETY
AND THE
CANADIAN INSTITUTE OF ACTUARIES



David L. Menning
Vice President Admissions

Daniel G. Roth
Chairperson
Examination Committee

Steven D. Armstrong
Assistant Chairperson
Examination Committee

May 3, 2011

Examination Committee

General Officers

Steven Armstrong

Arthur Placek

Jason Russ

Thomas Struppeck

Rhonda Port Walker

Geoffrey Werner

William Wilder

Arlene Woodruff

4 HOURS

Exam 7

Estimation of Policy Liabilities, Insurance
Company Valuation, and ERM

INSTRUCTIONS TO CANDIDATES

1. This 67.75 point examination consists of 22 problem and essay questions.
2. For the problem and essay questions the number of points for each full question or part of a question is indicated at the beginning of the question or part. Answer the questions on the lined sheets provided in your Examination Envelope. Use dark pencil or ink. Do not use multiple colors.
 - Write your Candidate ID number and the examination number, 7, at the top of each answer sheet. Your name, or any other identifying mark, must not appear.
 - Do not answer more than one question on a single sheet of paper. Write only on the front lined side of the paper -- **DO NOT WRITE ON THE BACK OF THE PAPER.** Be careful to give the number of the question you are answering on each sheet. If your response cannot be confined to one page, please use additional sheets of paper as necessary. Clearly mark the question number on each page of the response in addition to using a label such as "Page 1 of 2" on the first sheet of paper and then "Page 2 of 2" on the second sheet of paper.
 - The answer should be concise and confined to the question as posed. When a specific number of items is requested, do not offer more items than the number requested. For example, if three items are requested, only the first three responses will be graded.
 - **In order to receive full credit or to maximize partial credit on mathematical and computational questions, you must clearly outline your approach in either verbal or mathematical form, showing calculations where necessary. Also, you must clearly specify any additional assumptions you have made to answer the question.**
3. Do all problems until you reach the last page of the examination where "END OF EXAMINATION" is marked.
4. Prior to the start of the exam you will have a **fifteen-minute reading period** in which you can silently read the questions and check the exam booklet for missing or defective pages. A chart indicating the point value for each question is attached to the back of the examination. Writing will NOT be permitted during this time and you will not be permitted to hold pens or pencils. You will also not be allowed to use calculators. The supervisor has additional exams for those candidates who have defective exam booklets.

CONTINUE TO NEXT PAGE OF INSTRUCTIONS

©2011 Casualty Actuarial Society

5. Your Examination Envelope is pre-labeled with your Candidate ID number, name, exam number, and test center. Do not remove this label. Keep a record of your Candidate ID number for future inquiries regarding this exam.
6. Candidates must remain in the examination center until two hours after the start of the examination. The examination starts after the reading period is complete. You may leave the examination room to use the restroom with permission from the supervisor. To avoid excessive noise during the end of the examination, candidates may not leave the exam room during the last fifteen minutes of the examination.
7. At the end of the examination, place all answer sheets in the Examination Envelope. Please insert your answer sheets in your envelope in question number order. Insert a numbered page for each question, even if you have not attempted to answer that question. Anything written in the examination booklet will not be graded. Only the answer sheets will be graded. Also place any included reference materials in the Examination Envelope. BEFORE YOU TURN THE EXAMINATION ENVELOPE IN TO THE SUPERVISOR, BE SURE TO SIGN IT IN THE SPACE PROVIDED ABOVE THE CUT-OUT WINDOW.
8. If you have brought a self-addressed, stamped envelope, you may put the examination booklet and scrap paper inside and submit it separately to the supervisor. It will be mailed to you. Do not put the self-addressed stamped envelope inside the Examination Envelope.

If you do not have a self-addressed, stamped envelope, please place the examination booklet in the Examination Envelope and seal the envelope. You may not take it with you. Do not put scrap paper in the Examination Envelope. The supervisor will collect your scrap paper.

Candidates may obtain a copy of the examination from the CAS Web Site.

All extra answer sheets, scrap paper, etc., must be returned to the supervisor for disposal.
9. Candidates must not give or receive assistance of any kind during the examination. Any cheating, any attempt to cheat, assisting others to cheat, or participating therein, or other improper conduct will result in the Casualty Actuarial Society and the Canadian Institute of Actuaries disqualifying the candidate's paper, and such other disciplinary action as may be deemed appropriate within the guidelines of the CAS Policy on Examination Discipline.
10. The exam survey is available on the CAS Web Site in the "Admissions/Exams" section. Please submit your survey by May 23, 2011.

END OF INSTRUCTIONS

EXAM 7 – SPRING 2011

1. (3 points)

Given the following information (\$000) for a line of business:

Accident	Written	Earned	<u>Cumulative Reported Losses</u>		
<u>Year</u>	<u>Premium</u>	<u>Premium</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>
2007	5,756	4,779	413	2,310	5,845
2008	6,907	5,735	0	541	1,309
2009	8,289	6,882	936	2,311	
2010	9,946	8,258	50		

- The tail factor from 36 months to ultimate is 1.050.

a. (1 point)

Use the least squares method to estimate ultimate losses for the 2009 accident year.

b. (0.5 point)

Discuss the reasonability of the estimate derived in part a. above, relative to the estimate that would be produced by the link ratio method.

c. (1.5 points)

Illustrate graphically the relationships between the link ratio method, budgeted loss method and least squares method in modeling the loss development process.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

2. (4 points)

Given the following loss reserving information as of December 31, 2010:

Accident <u>Year</u>	On-Level Earned <u>Premium</u>	Growth <u>Function</u>	Reported <u>Losses</u>
2008	\$13,500	78.9%	\$7,200
2009	14,000	57.9%	5,700
<u>2010</u>	<u>14,500</u>	13.8%	<u>1,400</u>
Total	42,000		14,300

- Parameter standard deviation for the total estimated unpaid claims is 796.
- The expected accident year loss emergence pattern (growth function) can be approximated by a loglogistic function of the form:

$$G(x | \omega, \theta) = x^\omega / (x^\omega + \theta^\omega),$$

where x denotes time in months from the average accident date to the evaluation date, and G is the growth function describing cumulative percent reported.

- The maximum likelihood estimates of the parameters are:

$$\omega = 1.956 \text{ and } \theta = 15.286.$$

- The actual incremental loss emergence follows an over-dispersed Poisson distribution with scaling factor $\sigma^2 = 9$.

a. (3 points)

Using a truncation point of five years, estimate the total unpaid claims using the Cape Cod method.

b. (1 point)

Calculate the coefficient of variation of the total unpaid claims estimated in part a. above.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

3. (4 points)

One implicit assumption underlying the chain ladder loss development method is independence of the accident years. Given the following loss development triangle and link ratios:

Accident <u>Year</u>	<u>Reported Losses (\$000)</u>				
	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>	<u>60 Months</u>
2006	3,100	4,185	4,813	5,053	5,104
2007	3,050	3,660	4,282	4,368	
2008	2,800	3,640	4,077		
2009	2,500	3,125			
2010	2,357				

	<u>Age-to-Age Loss Development Factors</u>			
	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>
2006	1.35	1.15	1.05	1.01
2007	1.20	1.17	1.02	
2008	1.30	1.12		
2009	1.25			

a. (3 points)

Using a 95% confidence interval, test the null-hypothesis that this triangle does not have significant calendar year effects.

b. (1 point)

Describe two other implicit assumptions underlying the chain ladder loss development method.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

4. (2 points)

Given the following loss development triangle:

<u>Year</u>	<u>Cumulative Reported Losses (\$000)</u>									
	<u>Age in Years</u>									
	<u>0</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>
0	5,400	10,800	17,280	24,192	29,030	33,385	36,056	37,498	37,873	37,873
1	5,800	12,039	19,898	28,773	37,369	46,089	54,105	58,895	64,579	
2	3,888	7,582	12,199	16,834	20,933	24,367	27,335	29,207		
3	4,900	9,930	16,062	22,669	27,533	32,092	35,391			
4	5,655	10,631	17,123	23,802	29,320	33,980				
5	4,950	10,131	16,568	23,712	29,215					
6	6,650	13,475	22,073	30,902						
7	4,890	10,269	16,512							
8	5,780	11,502								
9	4,964									

A linear regression of the incremental losses against the previous cumulative losses was performed yielding the following table:

	<u>0 to 1</u>	<u>1 to 2</u>	<u>2 to 3</u>	<u>3 to 4</u>	<u>4 to 5</u>	<u>5 to 6</u>
Constant a	-66	-733	-740	-2,638	-4,183	-4,247
Standard Dev. of a	920	442	960	2,494	2,305	3,195
Factor b	1.024	0.692	0.453	0.352	0.323	0.250
Standard Dev. of b	0.171	0.041	0.055	0.106	0.079	0.092

a. (1 point)

Use the data above to demonstrate whether the chain ladder method or the linear plus a constant method better models the emergence of losses.

b. (1 point)

Support graphically the answer from part a. above.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

5. (1.5 points)

Given the following data for a retrospectively rated policy:

Retro Adjustment Period	<u>For Losses Capped at Per Accident Limit</u>		Loss Elimination Ratio from Per Accident Limit
	<u>Insurance Charge at Retro Maximum</u>	<u>Insurance Charge at Retro Minimum</u>	
First	0.113	0.004	4.5%
Second	0.142	0.003	5.5%
Third	0.154	0.003	6.2%
Subsequent	0.158	0.002	6.5%

a. (1 point)

Calculate the cumulative loss capping ratios for each retrospective adjustment period.

b. (0.5 point)

Explain the trend in the loss capping ratios from part a. above.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

6. (2.75 points)

Given the following information for accident year 2009 workers compensation business with a \$250,000 deductible as of December 31, 2010:

	<u>Ground Up Reported Losses</u>	<u>Ground Up Reported Severity</u>	<u>24 Months to Ultimate LDF</u>
Limited to \$250,000	\$4,000,000	\$125,000	1.25
Unlimited	4,500,000	140,000	1.50

- Full coverage premium for all policies: \$12,500,000
- Full coverage expected loss ratio: 0.700
- Excess ratio (per-occurrence charge): 0.150
- Excess ratio (aggregate charge): 0.020

a. (1.25 points)

Estimate the IBNR for losses above the deductible using the direct development method.

b. (1.25 points)

Estimate the IBNR for losses above the deductible using the Bornhuetter-Ferguson method.

c. (0.25 point)

Briefly describe an advantage of using the Bornhuetter-Ferguson method to estimate IBNR for workers compensation high deductible business.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

7. (1.25 points)

Given the following cumulative reported loss emergence pattern:

<u>Accident Year Age in Months</u>			
<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
15%	35%	65%	80%

- The loss emergence follows an over-dispersed Poisson model, with dispersion parameter, ϕ , equal to 1.3.
- Expected ultimate losses for accident year 2009 are \$150,000,000.

a. (0.75 point)

Calculate the variance of the 24 to 36 month incremental reported losses for accident year 2009.

b. (0.5 point)

Explain why modeling paid losses or claim counts may be preferable to modeling reported losses when using an over-dispersed Poisson model.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

8. (4 points)

Given the following cumulative paid loss information:

Accident Year	<u>Age of Development in Months</u>				
	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>
2006	\$223,000	\$303,000	\$354,000	\$376,000	\$381,000
2007	236,000	298,000	365,000	399,000	
2008	305,000	345,000	390,000		
2009	260,000	333,000			
2010	311,000				

	<u>12-Ult</u>	<u>24-Ult</u>	<u>36-Ult</u>	<u>48-Ult</u>	<u>60-Ult</u>
Estimators of loss development factors	1.599	1.280	1.092	1.013	1.000
Estimators of σ_j^2	2,410	716	173	42	0

- The estimation variance of the unpaid claims for accident year 2009 as of December 31, 2010 is \$157 million.

Use Mack's model, as described by England and Verrall, to calculate the prediction error as a percentage of the unpaid claims for accident year 2009 as of December 31, 2010.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

9. (2.25 points)

Given the following triangle of cumulative paid loss data (\$000):

Accident	<u>Age of Development in Months</u>				
<u>Year</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>
2006	510	605	583	592	607
2007	543	613	550	654	
2008	552	526	535		
2009	477	630			
2010	509				

- Assume the dispersion parameter, ϕ , is 1.25 for the negative binomial model.

a. (1.25 points)

Use the negative binomial model to calculate the mean and variance of $C_{2,2}$, the incremental claim amount for accident year 2007 from 12 to 24 months.

b. (0.75 point)

Explain why it is not appropriate to fit a negative binomial model to the full triangle above.

c. (0.25 point)

Identify a model that is appropriate to fit to the full triangle which would reproduce the chain ladder unpaid claim estimate.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

10. (3.25 points)

Given the following information for a line of business for a reinsurance company as of December 31, 2010:

Calendar Accident Year	Earned Risk Pure Premium (\$000)	Adjusted Premium (\$000)	Aggregate Reported Loss (\$000)	Aggregate Loss Report Lag	Chain Ladder IBNR (\$000)
2005	9,200	10,000	9,500	95%	500
2006	10,100	12,000	8,000	85%	1,412
2007	11,500	11,500	7,300	72%	2,839
2008	12,300	13,000	6,500	55%	5,318
2009	13,500	14,100	4,800	40%	7,200
<u>2010</u>	<u>15,000</u>	<u>15,000</u>	<u>2,500</u>	<u>28%</u>	<u>6,429</u>
Total	71,600	75,600	38,600		23,698

a. (1.5 points)

Use the Stanard-Bühlmann method to estimate the total IBNR.

b. (0.5 point)

Briefly describe one advantage and one disadvantage of using the Stanard-Bühlmann method to calculate IBNR for a reinsurer.

c. (1.25 points)

Use a credibility-weighted combination of the Stanard-Bühlmann and chain ladder estimates with a linear function of the report lag as the credibility weight to calculate a total IBNR estimate. For the chain ladder estimate, use a credibility factor of 0.6.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

11. (3.75 points)

a. (1.5 points)

Describe three reasons why claim report lags for reinsurers tend to be longer than for primary insurers.

b. (1 point)

Describe two reasons why reported losses for reinsurers generally develop more than for primary insurers after the claims have been reported.

c. (0.75 point)

Briefly describe three reasons why many standard actuarial loss development methods used by primary insurers may not work as well for reinsurers.

d. (0.5 point)

Briefly describe two reasons why industry loss development statistics may not be useful for estimating loss development patterns for a reinsurer.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

12. (3.25 points)

Given the following information to be used in the valuation of a firm:

Firm's ROE:	15%
Firm's plowback ratio:	90%
Market return:	9%
Yield on long-term T-bonds	5.2%
Liquidity premium for long-term T-bonds:	1.2%
Average growth rate for firms in the same industry:	4.2%
Average Beta for firms in the same industry:	1.4
Beta for this firm:	2.0

- The above values are assumed to continue indefinitely into the future.

Projections for the firm's net income after tax:

<u>Calendar Year</u>	<u>Net Income after Tax (\$000)</u>
2011	\$10,100
2012	11,464
2013	13,011

a. (0.75 point)

Demonstrate why it is reasonable for this firm's Beta to be significantly higher than the average Beta for firms in its industry.

b. (2 points)

Use the dividend discount model to estimate the value of this firm as of December 31, 2010.

c. (0.5 point)

Discuss how a company's growth rate may affect its dividend payout rate and risk-adjusted discount rate.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

13. (1.25 points)

Given the following information for a firm:

Return on equity (ROE):	10%
Book value growth rate:	5%
Discount rate:	8%

- The above values are projected to continue indefinitely.
- As of December 31, 2010, the firm's book value was \$1,000,000.

a. (0.75 point)

Use the price-to-book value ratio to estimate the value of the firm as of December 31, 2010.

b. (0.5 point)

Discuss whether it is reasonable to assume that the relationships among the values for the ROE, the book value growth rate and the discount rate are likely to continue indefinitely.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

14. (2 points)

A firm is incorporating extreme events into an enterprise risk model. Discuss the difficulty in producing accurate results for the model when considering:

a. (0.5 point)

Catastrophe risks.

b. (0.5 point)

Operational risk loss data.

c. (0.5 point)

Expert opinion.

d. (0.5 point)

The causes of extreme events.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

15. (2 points)

An insurance carrier has an internal model that determines its required economic capital.

a. (1 point)

Explain how the carrier's economic capital may not equal its:

- i. Rating agency capital.
- ii. Economic impairment earnings level.

b. (0.5 point)

Explain how the carrier's internal model might be changed in order to assess the economic impairment earnings level.

c. (0.5 point)

Briefly describe two situations where a single internal model may not be practical to use for multiple capital and ERM applications.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

16. (2.5 points)

An insurance carrier wants to allocate capital by line of business using an appropriate risk measure to do so. The carrier writes two lines as follows:

- Line A is subject to catastrophe losses.
- Line B is excess casualty.

a. (1 point)

For each line, provide a moment-based or tail-based risk measure that would not be appropriate to use to allocate capital. Explain your answer.

b. (1 point)

For each line, provide a moment-based or tail-based risk measure that would be appropriate to use to allocate capital. Explain your answer.

c. (0.5 point)

Explain whether a distortion measure for Line A would make more sense than a moment-based or tail-based risk measure when allocating capital.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

17. (3 points)

An insurance company that writes catastrophe coverage is considering the use of reinsurance in 2012 in order to reduce risk.

Expected premium, loss and expense for 2012 are as follows:

- Expected gross premium: \$500 million
- Expected gross losses: \$150 million
- Expected gross expenses: \$60 million

Three possible reinsurance options are being considered as follows:

	<u>Option 1</u>	<u>Option 2</u>	<u>Option 3</u>
Expected reinsurance premium	No	\$250 million	\$100 million
Expected reinsurance recoveries	Reinsurance	\$80 million	\$30 million

Assume no reinsurance commission expense and the cost of capital is 12%.

The distribution of 2012 modeled underwriting profit is partially described as follows (\$000,000):

<u>Percentile</u>	<u>Gross</u>	<u>Option 2</u>	<u>Option 3</u>
90.00%	250	(15)	100
95.00%	(10)	(50)	(200)
99.00%	(1,400)	(300)	(400)
99.50%	(2,200)	(420)	(600)
99.90%	(4,500)	(2,000)	(2,500)

a. (0.75 point)

Use an expected value basis to determine which option the company should select.

b. (2.25 points)

The company requires capital equal to 1.5 times the 1-in-200VaR. Based on the allocated risk capital methodology, determine the savings in capital cost of both options 2 and 3.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

18. (2.5 points)

An insurer that writes only commercial property insurance wants to model its underwriting risk. The carrier currently writes in only five states.

Over the next five years the carrier plans to:

- Begin writing commercial auto, excess umbrella and workers compensation insurance.
- Expand into more states.
- Sell more insurance directly through the internet, thereby eliminating some broker commissions.
- Significantly increase loss control efforts to mitigate losses with a corresponding increase in expenses.

a. (1 point)

For each of the four planned expansions, briefly discuss how the firm can model its underwriting risk using best practices.

b. (0.5 point)

Describe a risk correlation that should be reflected in the modeling of underwriting risk for this firm.

c. (0.5 point)

Describe the impact of not reflecting the correlation identified in part b. above in the model.

d. (0.5 point)

Describe a latent risk that should be reflected in the modeling of underwriting risk for this firm.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

19. (4 points)

The Chief Risk Officer for an insurance company is evaluating various risk measures for the purpose of allocating economic capital to the two business units within the company and has computed the following at the 99% confidence level:

	<u>Business Unit 1</u>	<u>Business Unit 2</u>	<u>Total</u>
Value at Risk (VaR)	\$100,000,000	\$90,000,000	\$200,000,000
Tail Value at Risk (TVaR)	105,000,000	160,000,000	290,000,000

a. (1.5 points)

Compare in detail the relative merits of Value at Risk (VaR) versus Tail Value at Risk (TVaR).

b. (1.5 points)

Calculate the Expected Policyholder Deficit for each business unit and for the company in total.

c. (1 point)

Fully discuss one possible method to allocate economic risk capital to each business unit, including reference to a potential problem with that method.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

20. (5.5 points)

An actuary is modeling a cat-exposed loss portfolio covering two lines of business and has the following loss history.

<u>Year</u>	<u>Loss</u>	
	<u>Line 1</u>	<u>Line 2</u>
2005	\$33,000,000	\$23,000,000
2006	28,000,000	22,000,000
2007	31,000,000	26,000,000
2008	27,000,000	21,000,000
2009	26,000,000	12,000,000
2010	22,000,000	20,000,000

a. (3 points)

Fully describe and calculate two statistical measures that can be used to quantify the dependence between these lines of business based on the loss history above.

b. (1 point)

Explain which of the two measures described in part a. above would better describe the dependence between these lines of business. Use the properties of these measures to support your answer.

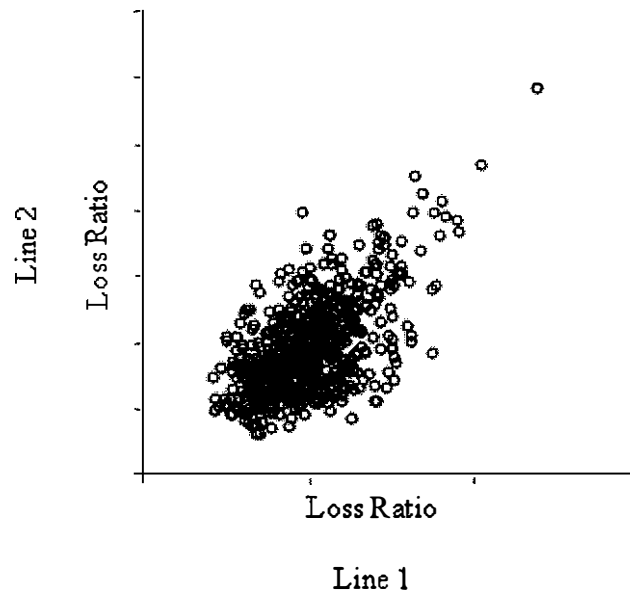
QUESTION 20 CONTINUED ON NEXT PAGE

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

(20 cont.)

There is also available a scatter-plot that represents the relationship between these two lines of business for the entire industry for 2005 through 2010. For each company in the industry, a point represents a pair of loss ratios for a given year.



c. (1 point)

Based on the scatter-plot above, identify a copula that would be appropriate to model the relationship between these two lines of business. Fully explain your answer.

d. (0.5 point)

Discuss a possible effect on the various tail-based risk measures of not selecting a copula that properly models the relationship between these two lines of business.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

21. (4 points)

Given the following information for an insurance company:

- Current firm value is \$950 million.
- Underwriting income for the year ended December 31, 2010 was \$40 million.
- Invested assets of \$200 million yielded a total return of 5% in 2010.
- Surplus at December 31, 2010 was \$130 million.
- Risk free interest rate is 3% per annum.

a. (1.5 points)

Calculate the probability of the firm entering a distressed state.

b. (1.5 points)

Discuss three issues that motivate insurers to adopt risk management to avoid financial distress.

c. (1 point)

Discuss two reasons it is difficult for distressed firms to raise bonds.

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2011

22. (6 points)

The Chief Financial Officer of a publicly traded insurance company entered into a questionable financial reinsurance agreement to avoid recognizing a sizable loss that came from the company's fastest growing business unit. The company's auditing committee failed to identify the transaction. When the news of the ongoing investigation by regulators became public, the company's stock price tumbled and it was reported that there might be a class action suit pending. Records also revealed that the Chief Financial Officer's compensation package included a large number of stock options that were redeemable at the end of the year.

a. (2.25 points)

Identify and describe three operational risks that were present in this situation.

b. (2.5 points)

Outline and briefly discuss the five steps suggested by Mango and Venter for managing operational risk.

c. (1.25 points).

Briefly discuss how well the five steps from part b. above would have prevented or mitigated this situation.

END OF EXAMINATION

Exam 7

Estimation of Policy Liabilities, Insurance Company Valuation, and ERM

May 3, 2011

POINT VALUE OF QUESTIONS

QUESTION	VALUE OF QUESTION	SUB-PART OF QUESTION						
		(a)	(b)	(c)	(d)	(e)	(f)	(g)
1	3.00	1.00	0.50	1.50				
2	4.00	3.00	1.00					
3	4.00	3.00	1.00					
4	2.00	1.00	1.00					
5	1.50	1.00	0.50					
6	2.75	1.25	1.25	0.25				
7	1.25	0.75	0.50					
8	4.00							
9	2.25	1.25	0.75	0.25				
10	3.25	1.50	0.50	1.25				
11	3.75	1.50	1.00	0.75	0.50			
12	3.25	0.75	2.00	0.50				
13	1.25	0.75	0.50					
14	2.00	0.50	0.50	0.50	0.50			
15	2.00	1.00	0.50	0.50				
16	2.50	1.00	1.00	0.50				
17	3.00	0.75	2.25					
18	2.50	1.00	0.50	0.50	0.50			
19	4.00	1.50	1.50	1.00				
20	5.50	3.00	1.00	1.00	0.50			
21	4.00	1.50	1.50	1.00				
22	6.00	2.25	2.50	1.25				
TOTAL	67.75							

Note regarding Sample Solutions for Exam 7, Spring 2011:

Due to the low number of candidates for this sitting only one model solution is provided for each question. Traditionally, model solutions are based on actual candidate responses. In cases where there was no full credit response for a particular question the model solution was either supplemented or replaced with the Exam Committee model answer.

Sample Solutions
Exam 07, 2011, Question 1

Solution 1

a)

Accident Year	Loss Ratio			
	12 months	24 months	36 months	ult
2007	0.086	2310/4779=0.483	5845/4779=1.223	1.223×1.05=1.284
2008	0.000	541/5735=0.094	1309/5735=0.228	0.228×1.05=0.240
2009	0.136	0.336		
2010	0.006			

$$\bar{x}=0.289$$

$$\bar{y}=0.762$$

$$\overline{x^2}=0.121$$

$$\overline{xy}=0.322$$

$$b = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2} = 2.685$$

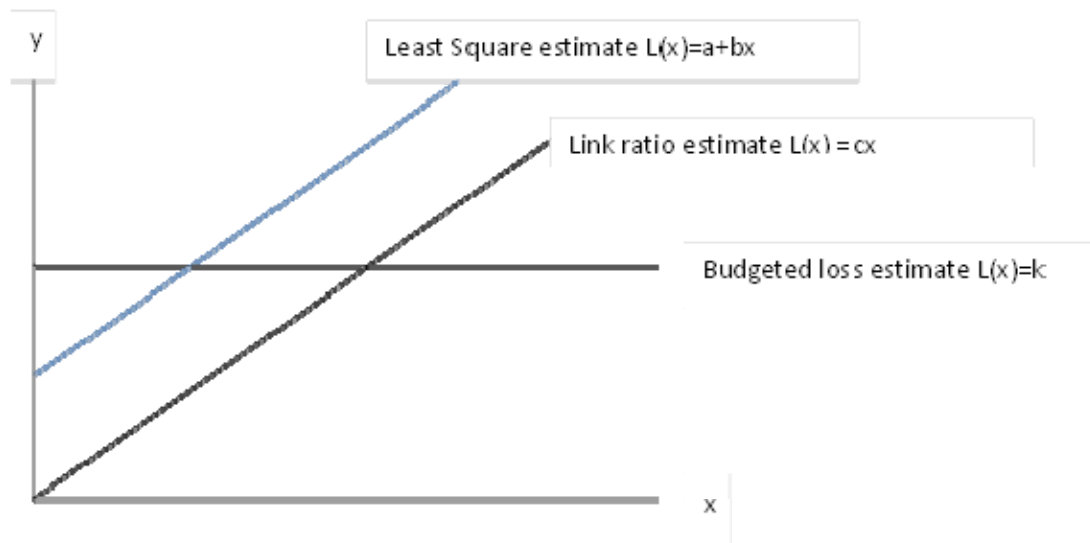
$$a = \bar{y} - b\bar{x} = -0.013$$

$$L(x) = -0.013 + 2.68 \times (2311/6882) = 0.89$$

$$\text{Ult loss} = 0.89 \times 6882 = 6111$$

b) In part a, a is less than zero. So our estimate of y will be negative for small value of x. That does not make sense. One might substitute the link ratio method while $a < 0$. Link ratio will produce positive number even for small values of x.

c)



Sample Solutions
Exam 07, 2011, Question 2

Solution 1

a)

	(1)	(2)	(3)	(4)	(5) = (1)×(4)	(6)	ELR = (6)/(5)
AY	On Level Earned Premium	Age at 12/31/2010	Average Age	Growth		Reported Losses	
2008	13,500	36	30	78.9%	10,652	7200	
2009	14,000	24	18	57.9%	8,106	5700	
2010	14,500	12	6	13.8%	2,001	1400	
	42,000				20,759	14,300	68.9%

Growth function for truncation = $x^{\omega} / (x^{\omega} + \Theta^{\omega}) = 54^{1.956} / (54^{1.956} + 15.286^{1.956}) = .9219$

(8) = 0.922 – Est Reserves = (1)
growth × ELR × (8)

13.3%	1,237
34.3%	3,309
78.4%	7,833
	12,378

b) No candidate supplied model answer – from grader model solution.

Process variance = $\text{Sigma}^2 \times E[R] = 9 \times 12,378 = 111,402$

Total Variance = Process Variance + Parameter Variance = $111,402 + 796^2 = 745,018$

Std Dev (Reserves) = $(745,018)^{0.5} = 86$

Sample Solutions
Exam 07, 2011, Question 3

Solution 1

a)

95% CI: statistic = 1.96

$$\begin{array}{rcl} \text{Median} = & \frac{12-24}{(1.25+1.3)/2} & \frac{24-36}{1.15} & \frac{36-48}{(1.02+1.05)/2} \\ = & 1.275 & 1.15 & 1.035 \end{array}$$

For development factors greater than median, its given "L", otherwise, given "S"

AY	12 - 24	24 - 36	36 - 48
06	L	×	L
07	S	L	S
08	L	S	
09	S		

$$E(Z_j) = n/2 - \binom{n-1}{m} \times (n/2^n)$$

$$\text{Var}(Z_j) = n \times (n-1)/4 - \binom{n-1}{m} \times (n \times (n-1)/2^n) + E(Z) - E(Z)^2$$

$$n = S + L$$

$$m = (n-1)/2$$

$$Z = \min(S, L)$$

Let j = Diagonal

J	N	m	Z	E(Z)	Var(Z)
1	1	0	0	0	0
2	1	0	0	0	0
3	3	1	0	0.75	0.1875
<u>4</u>	<u>3</u>	<u>1</u>	<u>0</u>	<u>0.75</u>	<u>0.1875</u>
			0	1.50	0.375

Null hypothesis = that there are no significant calendar year effects

Confidence interval: $1.5 \pm 1.96 \times \sqrt{0.375} \Rightarrow (0.275, 2.725)$

Since $Z=0$ is not contained in the interval, you reject the null hypothesis and suggest that there are significant calendar year effects

b)

Assumption 1 - the future incremental expected loss is proportional to the losses cumulative to date

Assumption 2 - the variance of the next incremental loss is a function of the age and cumulative losses to date

Sample Solutions
Exam 07, 2011, Question 4

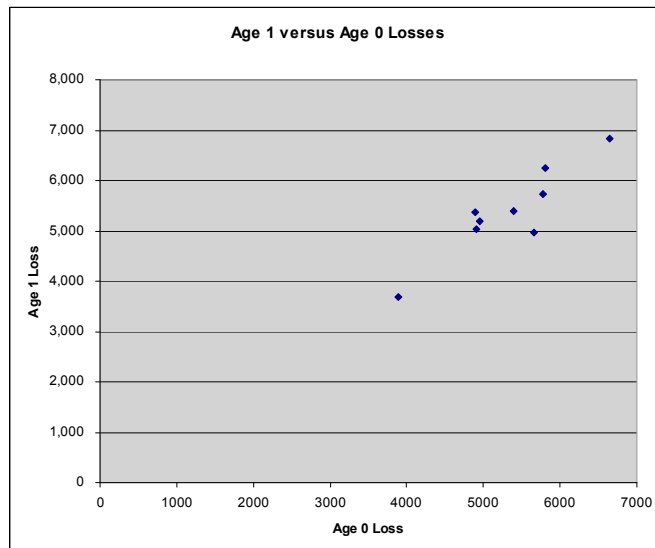
Solution 1

a.)

Significance of Factors

Usually the absolute value of a factor is required to be at least twice its standard deviation for the factor to be significant. So the table suggests that the b factors for the chain ladder method are significant. The chain ladder model is thus supported by this data.

- b.) One can plot Age d+1 incremental losses against Age d cumulatives. The chain ladder results should be something close to a straight line through the origin with slope equal to the development factor.



A factor by itself would be a straight line through the origin with slope equal to development factor. The above chart suggests the chain ladder is a better model.

Sample Solutions
Exam 07, 2011, Question 5

Solution 1

a)

Retro Adjusted Period (1)	Insurance Charge at Retro Max (2)	Insurance Charge at Retro Min (3)	Eliminated by Retro Max/Min (4) = (2)-(3)	LER from per Accident Limit (5)	Cumulative Loss Capping Ratio (6)
1	.113	.004	.109	.045	.846
2	.142	.003	.139	.055	.806
3	.154	.003	.151	.062	.787
3+	.158	.002	.156	.065	.779

b)

The loss capping ratios given in the problem are decreasing. Over time an increasing proportion of the loss development occurs outside of the loss limitation, lowering the capped losses as a percent of the total loss.

Sample Solutions
Exam 07, 2011, Question 6

Solution 1

a)

Excess losses = Unlimited losses – Limited losses = \$4.5 M - \$4.0 M = \$0.5 M

Find Excess LDF : Unlimited loss x Unlimited LDF = Limited loss x Limited LDF +

Excess loss × Excess LDF

\$4.5 M x 1.5 = \$4 M × 1.25 + \$0.5 M × Excess LDF

Excess LDF = 3.5

Excess IBNR = (Reported Excess losses) × (Excess LDF -1) = \$0.5 M × (3.5 – 1) = \$1.25 M

b)

$IBNR_{BF} = \text{Expected loss} \times \text{unreported \%} ; \text{unreported \%} \text{ xs loss}$

$= 1 - 1/3.583 = 72.093\%$

Expected xs loss = $12.5 \text{ M} \times .7 \times .15 + 12.5 \times .7 \times (1-.15) \times .02 = 1.46125 \text{ M}$

$IBNR_{BF} = \text{Expected xs loss} \times (1 - 1/3.583) = 1.46125 \text{ M} \times .72093 = 1,053,459$

c) Advantage of B/F method -> is more stable overall and can be tied to pricing estimates

Sample Solutions
Exam 07, 2011, Question 7

Solution 1

a)

Expected Ultimate Loss = 150,000,000

2009 24-36 development = .65 - .35 = .3

Expected 24-36 development = 150,000,000 $\times$.3 = 45,000,000

Over Dispersed Poisson Variance = Dispersion factor $\times$ Mean = 1.3 $\times$ 45,000,000
=58,500,000

b)

Paid losses and claim counts are not as susceptible to having negative incremental development. Paid losses and claim counts are not affected by changes in the case reserve estimates that affect incurred losses. A limitation of the over dispersed Poisson distribution is the sum of incremental claims in column j must also be positive.

Sample Solutions
Exam 07, 2011, Question 8

Solution 1

a) Process variance equation for accident year(i) reserves¹

$$\text{Var}[R_i] \approx \hat{D}_{in}^2 \sum_{k=n-i+1}^{n-1} \frac{\hat{\sigma}_{k+1}^2}{\hat{\lambda}_{k+1}^2 \hat{D}_{ik}}$$

<ignoring the hats in the equations that follow, carrying decimal places from calculations>

$$D_{2009,5} = \text{Ultimate loss for AY 2009} = 333,000 \times 1.28 = 426,240$$

$$24-36 \text{ ldf} = 1.280/1.092 = 1.172$$

$$36-48 \text{ ldf} = 1.092/1.013 = 1.078$$

$$D_{2009,3} = D_{2009,2} \times (\text{ldf}_{24-36}) = 333,000 \times 1.172 = 390,330$$

$$D_{2009,4} = D_{2009,3} \times (\text{ldf}_{36-48}) = 390,330 \times 1.078 = 420,770$$

$$\text{Var}[R_{2009}] = 426,240^2 \times \left\{ \frac{173}{1.172^2 \times 333,000} + \frac{42}{1.078^2 \times 390,330} + \frac{0}{1.013^2 \times 420,770} \right\}$$

Therefore, process variance = 85,537,979

MSEP for AY2009 = process variance + prediction variance

$$\begin{aligned} &= 85,537,979 + 157,000,000 \\ &= 242,537,965 \end{aligned}$$

Prediction error = sqrt(MSEP) = 15,574

AY 2009 reserves = ultimate loss – current paid

$$\begin{aligned} &= 426,240 - 333,000 \\ &= 93,240 \end{aligned}$$

Prediction error as % of reserves for AY 2009 = 15,574 / 93,240 = 16.7%

Sample Solutions
Exam 07, 2011, Question 9

Solution 1

a)

$$E[C_{2,2}] = (\lambda_2 - 1)D_{2,1} = (1.140 - 1) \times 543,000 = \underline{76,156}$$

$$\text{Var}[C_{2,2}] = \phi \lambda_2 (\lambda_2 - 1) D_{2,1} = 1.25 \times 1.140 \times (1.140 - 1) \times 543,000 = \underline{108,545}$$

$$\lambda_2 = \text{Vol wgted 12-24 ldf} = (605+613+526+630)/(510+543+552+477) = 1.140$$

b)

$$\text{Vol wgted 24-36 ldf} = (583+550+535)/(605+613+526) = 0.956 < 1.000$$

The variance of the incremental claims is $\text{Var}[C_{ij}] = \phi \lambda_j (\lambda_j - 1) D_{i,j-1}$

If $\lambda_j < 1.000$, that implies a negative variance, which is not possible.

Or it's not appropriate to use negative binomial for full triangle because there are negative incremental claims existing in the triangle which implies a negative variance and is not possible.

c)

Normal approximation to the negative binomial model

Sample Solutions
Exam 07, 2011, Question 10

Solution 1

a)

$$\begin{aligned}\text{SB ELR} &= \text{Reported Losses} / \Sigma (\text{Adjusted Premium}(k) \times \text{Report Lag}(k)) \\ &= 38,600 / (10,000 \times 0.95 + 12,000 \times 0.85 + 11,500 \times 0.72 + 13,000 \times 0.55 + 14,100 \times 0.40 + \\ &\quad 15,000 \times 0.28) \\ &= 0.858\end{aligned}$$

$$\begin{aligned}\text{SB IBNR} &= \text{SB ELR} \times \Sigma (\text{Adjusted Premium}(k) \times (1 - \text{Report Lag}(k))) \\ &= 0.858 \times (10,000 \times 0.05 + 12,000 \times 0.15 + 11,500 \times 0.28 + 13,000 \times 0.45 + 14,100 \times 0.60 + \\ &\quad 15,000 \times 0.72) \\ &= \mathbf{26,291}\end{aligned}$$

b)

SB method advantage: use actual claims data to derive ELR instead of judgmentally picking it like the BF.

SB method disadvantage: It's hard/difficult to obtain accurate adjusted on-level premium data.

c)

$$\begin{aligned}\text{Credibility Weighted IBNR} &= \Sigma \text{Credibility Factor} \times \text{Report Lag}(k) \times \text{CL IBNR}(k) + (1 - \\ &\quad \text{Credibility Factor}) \times \text{Report Lag}(k) \times \text{SB IBNR}(k) \\ &= (0.6 \times 0.95 \times 500 + (1 - 0.6 \times 0.95) \times 429) + (0.6 \times 0.85 \times 1412 + (1 - 0.6 \times 0.85) \times 1,545) + \\ &\quad (0.6 \times 0.72 \times 2839 + (1 - 0.6 \times 0.72) \times 2,764) + (0.6 \times 0.55 \times 5318 + (1 - 0.6 \times 0.55) \times 5,021) + \\ &\quad (0.6 \times 0.40 \times 7200 + (1 - 0.6 \times 0.40) \times 7,262) + (0.6 \times 0.28 \times 6429 + (1 - 0.6 \times 0.28) \times 9,270) \\ &= \mathbf{25,903}\end{aligned}$$

Sample Solutions
Exam 07, 2011, Question 11

Solution 1

a)

1. Longer reporting pipeline (reported to cedant, to reinsurance department, to intermediary, to reinsurer)
2. May also be lengthened by undervaluation of serious claims by cedant.
3. Some mass tort claims have extreme delays in discovery; loss dates may be decided by courts.

b)

In general, economic and social inflation causes this development. There could be a tendency to underreserve allocated loss adjustment expenses. Thus, early on the available information may indicate that a claim will pierce the reinsurance retention, but not yet indicate the ultimate.

c)

1. The exposure assumed by a reinsurance company can be extremely heterogeneous. This is a problem because most loss reserving methods require homogeneous data.
2. Many reinsurance contracts are unique and even when there exists larger aggregates of similar exposure, claim frequency may be so low and report lags so long that there is extreme fluctuation in historical data.
3. A reinsurer knows much less about specific exposures being covered than does a primary carrier.

d)

1. Much less specificity of information
2. Longer claim report

Sample Solutions
Exam 07, 2011, Question 12

Solution 1

a)

Firm's Dividend Growth Rate = $g = \text{Plowback Ratio} \times \text{ROE} = 90\% \times 15\% = 13.5\%$
 The average industry growth rate is given as 4.2%

Since the firm's growth rate is significantly higher than the industry's you can surmise that this firm is riskier than the average firm in the industry and therefore should have a Beta greater than the industry average.

b)

Determine the discount rate to use (CAPM):

Risk Free Rate = Yield on T-Bonds - Liquidity Premium on T-Bonds = $5.2\% - 1.2\% = 4.0\%$

Discount Rate = $k = (\text{Risk Free Rate}) + \text{Beta} \times (\text{Equity Risk Premium-diff. between expected market return and risk free})$
 $= 4.0\% + 2.0 \times (9.0\% - 4.0\%) = 14.0\%$

Determine the dividends and terminal value, then discount to get the company valuation:

Div = Net Income $\times$ Plowback ratio

	$10,100 \times (1-90\%)$	
DIV2011:	=	1,010
DIV2012:	$11,464 \times 10\% =$	1,146
DIV2013:	$13,011 \times 10\% =$	1,301

$\text{Terminal Value} = \frac{E(\text{Dividend}_{2014}) \times (1+g)}{k-g} = \frac{(1301) \times (1+13.5\%)}{14.0\% - 13.5\%} = 295,377$

Value of Firm as of 12/31/10:

= $\text{Exp}(\text{Dividend 2011})/(1+k) + \text{Exp}(\text{Dividend 2012})/(1+k^2) + \text{Exp}(\text{dividend 2013})/(1+k^3)$
 + Terminal Value/($1+k^3$)
 $= 1,010/(1.14) + 1,146/(1.14^2) + 1,301/(1.14^3) + 295,377/(1.14^3)$
 $= 202,017$

c)

High dividend payout rates are unlikely to be sustainable and so the effects of high growth rates are likely to be offset by lower dividend amounts. The dividend payments for firms with high growth rates are likely to be riskier than those of firms with low growth rates. As a result, the effect of high growth rates are likely to be offset by discounting dividends to present value using higher risk adjusted discount rates.

Sample Solutions
Exam 07, 2011, Question 13

Solution 1

a)

$$\text{ROE} = 10\%$$

$$k = 8\%$$

$$\text{BV}_0 = 1,000,000$$

$$\begin{aligned} P / BV &= 1 + [(\text{ROE} - k) / (k - g)] \\ &= 1 + [(.10 - .08) / (.08 - .05)] \\ &= 1.667 \end{aligned}$$

$$P / BV \times BV = 1.667 \times 1,000,000 = 1,667,000$$

b)

It is not likely that the company will continue to see abnormal earnings indefinitely since as you continue to earn profit, this will attract more suppliers and prices will drop to remain competitive. This would lower the return until no abnormal earnings exist and returns are as expected. (Note that abnormal earnings is meant to represent earnings more than your discount rate, your expected return)

Sample Solutions
Exam 07, 2011, Question 14

Solution 1

a)

Catastrophes are very infrequent so few historical data points are available. Every future catastrophe will affect the model and its subsequent predictions, and there is considerable variability in possible loss amounts.

Any 2 of the following examples:

- The degree of detail and research that is put into separate cat models varies significantly so some models are more accurate than others.
 - The assumed frequency of occurrence of the modeled cat event is usually entirely based on judgment.
 - These models often draw on extremely sparse data to determine the potential magnitude of a catastrophic event.
 - After determining potential magnitude of event, cat models usually attempt to estimate the impact of such an event allowing for changes in conditions (such as national response plans, medical advances, changes in social habits, changes in building techniques etc). This involves judgment.
- b) Operational risk loss data is often taken from third party databases and may, for example, include extreme loss events experienced by other industries (the banking industry for example). These would then be scaled to a size appropriate to the insurer conducting the modeling by using reference metrics; this approach introduces its own risk. Other appropriate answers may be accepted.
Operational risk extreme events are also often developed solely based on discussion among experts within a business, who naturally draw on their own experiences and knowledge of the experiences of other similar companies. This is subject to the biases inherent in the use of judgment.
- c) Using experts within a business involves drawing on their own experiences and knowledge of other similar companies. This is subject to the problems inherent in the use of judgment -for example their knowledge may be about different types of firms, they may have limited information, their judgment may be biased, etc..
- d) The analysis undertaken, and the model design for extreme events, often focuses on the outcomes of a risk rather than the causes. The models generally make limited or no adjustment for the possible evolution of the risk environment such as
- increased globalization
 - growth of leverage
 - increased/decreased regulation/supervisory oversight

Sample Solutions

Exam 07, 2011, Question 15

Solution 1

- a)
 - i. This may be calculated using a separate rating agency factor-based model.
 - ii. This “earning at risk” measurement is often assessed within an internal model at a lower return period so it may not equal to economic capital.
 - iii. In order to avoid reaching an “economic impairment earning level” a different value of capital may be required.
- b)

Economic impairment earnings level measurement is often assessed within an internal model at a lower return period (e.g. the 90% level of sufficiency).
- c)
 - i. Day-to-day business decisions (such as product approvals) are difficult to make with company level internal model that has a long run time.
 - ii. Groups may have a variety of models in use where the group is managed on decentralized lines.
 - iii. allocating capital back to lines of business may not be practical using just one model.
 - iv. carrier’s overall company-level model may be simpler than needed for product pricing.

Sample Solutions
Exam 07, 2011, Question 16

Solution 1

a)

Line A – VAR would not be appropriate as it is not coherent (fails subadditivity). Also VAR is a single percentile of the loss distribution and does not describe the risk in the tail

Line B – Standard deviation would not be appropriate as it does not work for skewed loss distributions. Also, standard deviation considers upside and downside where for xs casualty, we're more concerned with downside.

b)

Line A – TVAR would be more appropriate as it describes the expected risk in the whole tail. TVAR is also a coherent risk measure.

Line B – RTVAR (TVAR + conditional standard deviation) risk in the tail increases for higher probabilities which is representative of the inherent uncertainty for extreme casualty losses. RTVAR is also non-linear for exceedance which is more reasonable for xs casualty where ALAE is more prevalent and less likely to be proportional to losses.

c)

A distortion measure for allocating capital would make sense for Line A as it shifts significant probability towards the tail and so gives consistent charges across the board to high-level catastrophe covers as well as to standard coverages, all with the same parameterization.

Sample Solutions

Exam 07, 2011, Question 17

Solution 1

a)

Gross - Option 1: UW Profit = $500 - 150 - 60 = 290$

Option 2: UW Profit = $500 - 150 - 60 - 250 + 80 = 120$

Option 3: UW Profit = $500 - 150 - 60 - 100 + 30 = 220$

Since the gross option results in the highest UW profit, option 1 should be selected by the company

b)

	Gross	Option 2	Option 3
Risk Capital Estimate	$1.5 \times 2,200 = 3,300$	$1.5 \times 420 = 630$	$1.5 \times 600 = 900$
Capital Cost	$.12 \times 3,300 = 396$	$.12 \times 630 = 75.6$	$.12 \times 900 = 108$
Savings in Capital Cost		$396 - 75.6 = \mathbf{320.4}$	$396 - 108 = \mathbf{288}$

Sample Solutions
Exam 07, 2011, Question 18

Solution 1

a)

For first two ventures, candidate could talk about pricing, claim issues and that model should account for:

- 1) Attritional claims – the risk that there is a higher frequency of small claims or that these claims have a higher average cost than expected.
- 2) Large claims – the risk that there are more large claims than expected or that the large claims received are more severe than expected.
- 3) Catastrophes – the risk that a catastrophe may occur causing higher claim frequency and/or severity than expected.
- 4) correlations, latent exposures.

For each of third and fourth expansions, candidate should recommend that model accounts for an expense or pricing issue such as:

- Is product priced to reflect changing expenses and/or market pressures.
 - Proportionate expenses such as commission should be modeled directly.
 - Profit commissions should be simulation specific.
 - Over time expense levels will be linked to inflation and the model should reflect this.
- There may be other acceptable answers.

b)

any of the following possible answers:

- There may be correlations between lines
- Attritional losses, particularly from longer tail business. These claims will often involve bodily injury and are sometimes subject to legislative change, whereby all cases settled after a particular date will be materially higher than previously experienced. This risk will correlate across years and business classes, affecting both underwriting and reserving risk. It is important that such “shocks” are modeled and that appropriate correlations are captured.
- Cross-class correlations that need to be captured. In particular a number of classes, otherwise uncorrelated, may exhibit some dependency in the tail of their distributions. Such dependency may not be contained in the available data so estimates of the level of the dependency will need to be chosen using considerable

c)

Answer should follow from what they selected in b and discuss result of not reflecting correlation of either double-counting an impact or underestimating one.

d)

Examples such as asbestos, other mass tort excess umbrella claims, comp occupational hazards etc., are acceptable. Candidate should also describe issues with estimating severity and likelihood of such an event happening.

Sample Solutions
Exam 07, 2011, Question 19

Solution 1

a)

VaR

Var is not sub additive as VaR at a fixed probability level for a number of risks combined can be greater than the sum of the VaRs of the individual risks. However for many applications, sub-additivity is not necessary.

It is just a single point on the distribution so does not give very much information.

It is also difficult to allocate reasonably to business units.

“TVaR is sometimes criticized for including losses excess of insolvency, which presumably do not affect shareholders. This is not really a problem, however, as these losses can hurt policyholders, whose attitudes affect growth and profits, and hence do affect shareholders. A more serious issue with TVaR is that it is linear for losses in the excess region. This is contrary to usual risk attitudes, where a loss twice as big is regarded as more than twice as bad.”

b)

$$EPD = (1 - \alpha) \times (TVaR - VaR)$$

Unit	α	TVaR	VaR	EPD
1	0.99	105,000,000	100,000,000	50,000
2	0.99	160,000,000	90,000,000	700,000
3	0.99	290,000,000	200,000,000	900,000

c)

VaR at a high level is computed for each unit, these are added up, and each unit's share of the total is taken to be its share of the economic capital. This is in part developed from the idea that firm capital is determined by a high VaR level, so that is the risk measure to use to allocate capital to each unit.

A problem with this method is that it is not marginal, and so the risk adjusted profit calculation does not maintain the financial principle of comparing marginal profits with marginal costs.

Sample Solutions
Exam 07, 2011, Question 20

Solution 1

a)

Kendall's τ – measures relative orders and does not take into account the magnitudes of values in the lines of business being compared. Weights are determined by the number of swaps.

Original:

	Line 1	Line 2	Rank 1	Rank 2
2005	33,000,000	23,000,000	1	2
2006	28,000,000	22,000,000	3	3
2007	31,000,000	26,000,000	2	1
2008	27,000,000	21,000,000	4	4
2009	26,000,000	12,000,000	5	6
2010	22,000,000	20,000,000	6	5

Reorder and swap

	Line 1(x)	Line 2(y)	Rank 1	Rank 2	Swap 1	Swap 2
2005	33,000,000	23,000,000	1	2	1	1
2007	31,000,000	26,000,000	2	1	2	2
2006	28,000,000	22,000,000	3	3	3	3
2008	27,000,000	21,000,000	4	4	4	4
2009	26,000,000	12,000,000	5	6	6	5
2010	22,000,000	20,000,000	6	5	5	6

Swaps: 1) switch 2007 and 2006 & 2) switch 2010 and 2009

Number of swaps $Q=2$

Number of elements $N=6$

$$\tau = 1 - [4Q/N \times (N - 1)]$$

$$\tau = 1 - [4(2)/6 \times (6 - 1)]$$

$$\tau = 1 - (8/30)$$

$$\tau = 0.7333$$

Pearson – a cardinal statistic weighting elements by the squares of their deviances from the mean.

Standard correlation on given values for Line 1 and Line 2.

$$\rho = \text{cov}(x,y) / [\text{std}(x) \times \text{std}(y)]$$

$$= [(xy\text{-bar}) - (x\text{-bar}) \times (y\text{-bar})] / [\text{std}(x) \times \text{std}(y)]$$

$$x\text{-bar} = 27,833,333$$

$$y\text{-bar} = 20,666,667$$

$$xy\text{-bar} = 583,333,333,333,333$$

$$\text{std}(x) = 3,531,603$$

$$\text{std}(y) = 4,307,616$$

$$\rho = \text{cov}(x,y) / [\text{std}(x) \times \text{std}(y)]$$

$$= (583,333,333,333,333 - 27,833,333 \times 20,666,667) / (3,531,603 \times 4,307,616)$$

$$= \mathbf{0.5332}$$

Spearman's rank correlation - measures relative orders and does not take into account the magnitudes of values in the lines of business being compared.

	Line 1 (x)	Line 2 (y)	Rank 1	Rank 2	di=rank 1-rank 2	di^2
2005	33,000,000	23,000,000	1	2	-1	1
2007	31,000,000	26,000,000	2	1	1	1
2006	28,000,000	22,000,000	3	3	0	0
2008	27,000,000	21,000,000	4	4	0	0
2009	26,000,000	12,000,000	5	6	-1	1
2010	22,000,000	20,000,000	6	5	1	1

Sum of difference² = 4

$$\text{Spearman's } \rho = 1 - \frac{\sum_{j=1}^n d_j^2}{\frac{N(N^2 - 1)}{6}}$$

$$\text{Spearman's } \rho = 1 - \frac{4}{\frac{6(36 - 1)}{6}} = \mathbf{0.8857}$$

- b) Both Kendall's Tau and Spearman's rank are ordinal measures, but Kendall's Tau has greater dependence in the tails. As the portfolio is exposed to Cats it is more reasonable to use Kendall's Tau to describe dependence.
Alternate: Ranks of preference for these measures is Kendall, Spearman, Pearson. Pearson is not well suited to measuring dependency in thick tailed, skewed distributions. Kendall is the same for a copula and any correlated variates produced by it.
- c) There is much more dependence in the right tail (almost completed dependence) than in the center of the loss ratio distributions. This suggests the Heavy Right Tail (HRT) copula which does exhibit considerable right-tail dependence.
Alternate: A copula that could capture the tail-dependency for high quartiles would be suitable. High loss ratio will happen seldom together but it seems as there is a positive correlation in the right tail and the average L/R are rather uncorrelated. I would suggest a Gumbel copula to be appropriate. The Gumbel copula can capture right-tail dependency.
- d) If you do not pick an accurate copula that measures the dependency in tails then your tail-based risk measures are underestimating the potential loss. As there is more correlation in tails, the more losses will be expected and at volatile amounts.

Sample Solutions
Exam 07, 2011, Question 21

Solution 1

a)

E = expected annual earnings

M = multiplier

E=UW Earnings + Investment Earnings

$$=40 + .05 \times 200=50$$

V = expected annual earnings $\times$ multiplier = 950

$$= E \times M = 50 \times M$$

$$V=950 \quad 950=50 \times M$$

$$M=19$$

$$19= D / (1-D) \quad 19-19D=D$$

$$19=20D \quad 19/20=0.95=D$$

The discount factor D reflects both time value of money and probability of distress.

$$D=(1-d) / (1+r)$$

$$=(1-d)/1.03$$

$$0.95= (1-d) / 1.03$$

$$0.95 \times 1.03 = 1-d$$

$$.9785= 1-d$$

$$d=.0215$$

b)

I – costs of financial distress – It can be difficult and costly for distressed firms to raise capital. Issuing Principal debtholders are the customers (policyholders)

Loss reserves and unearned premium reserves are supported by funds held for payment to shareholders

The above gives debtholders more relative power in the relationship.

Insurers must maintain lower levels of risk because the relationship with debt holders is ongoing and customer relationships will suffer if their risk attitudes are ignored.

I.

Insurers are particularly vulnerable to financial distress.

Insurance payment is a promise of payment contingent upon certain future events.

A distressed firm can lose many customers which may cause it to lower prices to attract business

Customers who remain expect discounted prices and this sets into motion a vicious cycle that includes substantial loss of future earnings potential.

II.

Reinsurance is the dominant means of hedging insurance risk

Since reinsurance is recorded in financial statements the insurance industry is more transparent than others in its use of risk transfer

c)

- I. Distressed firms are able to issue only high yield bonds
Which involve substantial recurring expense in terms of making high interest payments
- II. Also, bonds often have covenants that require specific firm action in the case of financial distress which restricts a firm's flexibility in responding to the distress.

Sample Solutions
Exam 07, 2011, Question 22

Solution 1

a)

3 of the following

- 1) Internal fraud – attempt to circumvent accounting regulations by misrepresenting the nature of the transaction.
- 2) HR risk – misdesign of compensation program – CFO is compensated largely by options that encourage risk-taking behavior. Employee incentives are misaligned with company goals.
- 3) People failure – improper use of reinsurance to conceal a sizable loss from a fast growing business unit.
- 4) Governance failure – internal controls should have prevented this improper act, but either were not in place or were ignored.
- 5) Reputational risk – Company reputation possibly irreparably damaged by bad press coverage following discovery of the act
- 6) Lawsuits – risk that shareholders will file suit against the company due to negligent controls in place to either prevent or identify improper internal activity.

b)

1. Identify exposure bases for each key operational risk source
2. Measure the exposure level for each business unit for each risk source
3. Estimate the loss potential (freq and sev) per unit of exposure for each operation risk
4. Combine 2. and 3. to produce modeled business unit loss freq and sev distribution
5. Estimate the impact of mitigation, process improvements or risk transfer on the business unit loss freq and/or sev dist.

c)

In this situation the fast growing business unit should have been monitored as its premium volume was increasing rapidly. Through monitoring it would have been revealed that the business unit's exposure was increasing too rapidly without adequate reinsurance. If this exposure was recognized controls could have been put in place to mitigate.